MESSI: A Nextflow pipeline for benchmarking multiomics integration methods for disease classification

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Background

Technological advances have allowed for the profiling of different molecular measurements (e.g., genes, proteins, metabolites) from the same biological samples; termed multiomics (Hasin *et al.*, 2017; Subramanian *et al.*, 2020).

The majority of multiomics data are obtained for samples from individual patients (bulk multiomics data) and more recently with single-cell and spatial resolution.

Many methods have been developed to jointly analyze multiomics data to identify common patterns between datasets, and to identify biomarkers of disease (Vandereyken *et al.*, 2023).

This integrative approach of multiomics data may strengthen the understanding of the molecular dynamics underlying the biological processes of diseases and may lead to novel strategies for early detection, prevention, and treatment of human diseases (Sun & Hu, 2016).

Researchers face difficulties in choosing the right method due to the varying nature of analytical tasks such as clustering analysis, factor analysis, or cancer type prediction.

Hence, numerous reviews have benchmarked various integrative methods (Bersanelli *et al.*, 2016; Cantini *et al.*, 2021; Hasin *et al.*, 2017; Huang *et al.*, 2017; Li *et al.*, 2018; Li *et al.*, 2022; Luecken *et al.*, 2022; Pucher *et al.*, 2019; Richardson *et al.*, 2016; Yu & Zeng, 2018; Zeng & Lumley, 2018). However, these studies often focus on methods for specific tasks or data types and may lack reproducibility.

To address this, we propose a new automated framework to systematically benchmark and analyze multiomics methods 1 with varying data types. This framework ensures full reproducibility and provides standardized input for any analytical task, with the capability to run on any computing platform.

Table 1: Overview of real datasets to benchmark

Method	Type	Language	Package available	Paper link	Code link
DIABLO	GCCA	R	yes	https://academic.oup.com/bioinformatics/article/35/17/3055/5292387	https://mixomicsteam.github.io/mixOmics-Vignette/id_06.html
MOFA+	Factor analysis	R/Python	yes	https://genomebiology.biomedcentral.com/articles/10.1186/s13059-020-02015-1	https://github.com/bioFAM/mofapy2/blob/master/mofapy2/notebooks/getting_started_python.ipynb
RGCCA	GCCA	R	yes	https://link.springer.com/article/10.1007/s11336-017-9573-x	https://github.com/rgcca-factory/RGCCA
IntegrAO	IntegrAo	Python	yes	https://www.nature.com/articles/s42256-024-00942-3	https://github.com/bowang-lab/IntegrAO/blob/main/tutorials/simulated_cancer_omics.ipynb
IntegratedLearner	IntegratedLearner	R	yes	https://onlinelibrary.wiley.com/doi/full/10.1002/sim.9953?saml_referrer	https://github.com/himelmallick/IntegratedLearner
Stabl	Stabl	Python	no	https://www.nature.com/articles/s41587-023-02033-x	https://github.com/gregbellan/Stabl/blob/stabl_lw/Notebook%20examples/Tutorial%20Notebook.ipynb
Cooperative Learning (multiview)	Penalized regression	R	yes	https://www.nature.com/articles/s41746-024-01128-2	https://github.com/dingdaisy/cooperative-learning/
MOGONET	GNN	Python	no	https://www.nature.com/articles/s41467-021-23774-w	https://github.com/txWang/MOGONET
GOAT	GNN	Python	no	https://academic.oup.com/bioinformatics/article/39/10/btad582/7280697	https://github.com/DabinJeong/GOAT2.0
mowgli	Matrix factorization, optimal transport	Python	yes	https://www.nature.com/articles/s41467-023-43019-2	https://github.com/cantinilab/Mowgli
BEAM	BEAM	R	yes	https://www.biorxiv.org/content/10.1101/2024.07.31.605805v1	https://annaseffernick.github.io/BEAMR/articles/BEAMR.html
SLIDE	SLIDE	R	yes	https://www.nature.com/articles/s41592-024-02175-z	https://github.com/jishnu-lab/SLIDE/blob/main/vignettes/SLIDE.pdf
JointNMF	JointNMF	JointNMF	JointNMF	JointNMF	https://cran.r-project.org/web/packages/nnTensor/vignettes/nnTensor-2.html
eipy	Ensemble	Python	yes	https://arxiv.org/abs/2401.09582	https://github.com/GauravPandeyLab/eipy
multiVI	Deep Learning	Python	yes	https://www.nature.com/articles/s41592-023-01909-9	https://docs.scvi-tools.org/en/stable/tutorials/notebooks/multimodal/MultiVI_tutorial.html

Results

MESSI pipeline

We introduce here the MESSI pipeline, *Multiple Experiments with SyStematic Interrogation* (Fig 1, and additional supplementary), which comprises of 4 main steps: 1) Prepare data, 2) Data splitting, 3) Cross validation, 4) Feature selection.

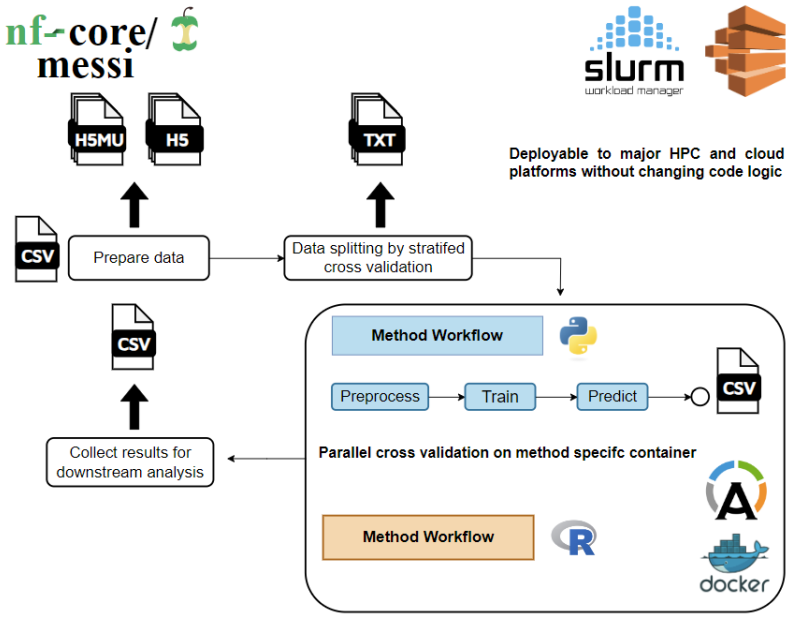

Figure 1: Workflow design of MESSI.

MESSI is implemented with Nextflow (Di Tommaso *et al.*, 2017), a domain specific language (DSL) primarily used by bioinformaticians. Compared to traditional workflow management systems like Snakemake (Köster & Rahmann, 2012) or GNU make (Stallman *et al.*, 1988), Nextflow breaks down complex workflows into modular components, and connect them with channels which determines the flow of pipeline. This feature allows us to customize each module, extending the pipeline to not only benchmark purposes but also multi-purpose usage. Additionally, This modular design enables rapid, efficient and reproducible way to test and maintain codebases.

Moreover, the core feature of Nextflow of using containerization for modules like Docker (Docker *et al.*, 2020) and Singularity (Kurtzer *et al.*, 2017) solves the reproducibility issue in replicating papers. Each time the pipeline is executed, containers are created for each process defined in the workflows that constitute the pipeline. These containers follow same sets of operating system configurations, software versions of the desired computation performed. Hence, it fulfills reproducibility and portability of our results as each time we run these independently on its own encapsulated environment. Furthermore, the independence between each process leads nature of parallelizable computations. This characteristic of the pipeline along with the resumability of re-executing interrupted or failed processes make debugging or changing computational settings for certain methods only atomic and simple. Ultimately, reducing time complexity of dealing with large datasets or long runtime computations, compared to standard way of sequentially executing complex scripts.

In terms of interoperability, Nextflow is capable to run the pipeline on various computing platforms including and not limited to our own personal computer, mainstream high-performance computing clusters (HPC) like SLURM, PBS or popular cloud platforms like AWS and Google cloud without modifying code logics. This is enabled from resource and parameters configurations, making user to only worry about configurations to carry out and the code logic itself.

Standardized data input in both R and Python

To ensure the standardized input, we incorporated the MultiAssayExperiment (Ramos *et al.*, 2017) package of the Bioconductor community (Gentleman *et al.*, 2004) for R (R Core Team *et al.*, 2013) and MuData (Bredikhin *et al.*,

2022) for Python (Sanner *et al.*, 1999). These tools are easily transferable, meaning users familiar with one can have the pipeline automatically convert to the other.

Currently, we are benchmarking existing methods like DIABLO (Singh *et al.*, 2019), Cooperative Learning (Ding *et al.*, 2022), MOGONET (Wang *et al.*, 2021), RGCCA (Girka *et al.*, 2023), MOFA (Argelaguet *et al.*, 2018, 2020) . The current pipeline focuses on testing this selection of integrative methods against a series of real multiomics datasets, particularly for classification tasks.

The results are compared using metrics like balanced classification error, F1 score, and biological enrichment results.

Through MESSI, we could systematically identify the top-performing methods with respect to classification performance and identifying molecular drivers of disease. Furthermore, this could extend to regression tasks or clustering tasks. This project can greatly benefit the community, either by developing new methods to assess their strength or finding a method that best suits custom multiomics data.

Prepare data

- Mention how data expect to be in tar gz containing the MAE and MuData
- These data goes to a preprocess step of filtering
- And cleaning up "columns" or metadata of it
- Describe some parameters used here, that could be controlled through nextflow config
- So from here, we come to commonly used and cleaned mae and mudata for respective language method

Data Splitting

- This splitting is because not all methods carry a built in cv
- Moreover, this make sure all method start with same data setup, since cv in each method might vary due to random seed problem
- Stratified to make sure of handling class imbalance, possible to go into discussion

Cross Validation

- Have a more detail flow of how here works, given could parallel per method, data, fold
- Show some plot from each method or data
- Then this makes each method as one workflow, where each can contain different number of modules, but requirement is just produce some kind of csv as end result that follows some format
- each workflow works in parallel and group by languages

Feature Selection

- This is an additional workflow that go directly after prepare data rather than splitting
- also makes it parallel by method
- then just talk how collect stuff in common format and produce series of metrics file that could be visualize downstream

Benchmark analysis

We applied the pipeline on the described methods upon the real datasets. We presented benchmark results for various multiomics data and methods at Fig **??** with real datasets from table 2.

```
## Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be
## resized.
```

Table 2: Overview of real datasets to benchmark

Dataset	N	Y=0	Y=1	Prop(Y = 1)	Omic	P	Disease
GSE38609	24	control Cer	autistic	0.458	mrna	8110	Autism
					cpg	1674	
					cc	12	
					mrna	5831	

Table 2: Overview of real datasets to benchmark *(continued)*

Dataset	N	Y=0	Y=1	Prop(Y = 1)	Omic	P	Disease
GSE71669	33	non-invasive bladder cancer	invasive bladder cancer	0.424	cpg	8915	Bladder Cancer
					cc	10	
ROSMAP	351	normal control	alzheimer's disease	0.519	epigenomics	58	Alzheimer's Dise
					genomics	74	
					transcriptomics	90	
TCGA-ACC	46	stagei	stageiii	0.391	Methylation_HM450K	7396	Adrenocortical C
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii		miRNA	166	
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii		RPPA	40	
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii		RNAseq_HiSeq	5660	
			stageiv				
		stageii	stageiii				
			stageiv				
TCGA-BLCA	336	stagei	stageiii	0.685	Methylation_HM450K	8072	Bladder Urotheli
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii		miRNA	268	
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii		RPPA	44	
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii		RNAseq_HiSeq	6093	
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii		Methylation_HM450K	7985	
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii				
			stageiv				

Table 2: Overview of real datasets to benchmark *(continued)*

Dataset	N	Y=0	Y=1	Prop(Y = 1)	Omic	P	Disease
TCGA-BRCA	109	stageii	stageiii	0.358	miRNA_GA	251	Breast Invasive C
			stageiv				
		stagei	stageiii		RPPA	43	
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii		RNAseq_HiSeq	5936	
			stageiv				
		stageii	stageiii				
			stageiv				
TCGA-CHOL	30	stagei	stageiii	0.233	Methylation_HM450K	7756	Cholangiocarcino
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii		miRNA	168	
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii		RPPA	48	
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii		RNAseq_HiSeq	5407	
			stageiv				
		stageii	stageiii				
			stageiv				
TCGA-ESCA	119	stagei	stageiii	0.353	Methylation_HM450K	7963	Esophageal Carc
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii		miRNA	202	
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii		RPPA	44	
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii		RNAseq_HiSeq	5709	
			stageiv				
		stageii	stageiii				
			stageiv				
			stageiii				

Table 2: Overview of real datasets to benchmark *(continued)*

Dataset	N	Y=0	Y=1	Prop(Y = 1)	Omic	P	Disease
TCGA-KICH	63	stagei	stageiv	0.302	Methylation_HM450K	6489	Adenomas and A
		stageii	stageiii				
			stageiv				
		stagei	stageiii		miRNA	180	
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii		RPPA	57	
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii		RNAseq_HiSeq	5517	
			stageiv				
		stageii	stageiii				
			stageiv				
TCGA-KIPAN		stagei	stageiii		Methylation_HM450K	7914	Kidney Cancer
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii		miRNA_GA_miR	215	
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii		RPPA	60	
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii		RNAseq_HiSeq	5951	
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii		Methylation_HM450K	7726	
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii		miRNA_GA_miR	214	
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii		RPPA	44	
			stageiv				
		stageii	stageiii				
			stageiv				

Table 2: Overview of real datasets to benchmark *(continued)*

Dataset	N	Y=0	Y=1	Prop(Y = 1)	Omic	P	Disease
							Kidney Renal Cl
		stagei	stageiii				
			stageiv				
		stageii	stageiii		RNAseq_HiSeq	5941	
			stageiv				
TCGA-MESO	63	stagei	stageiii	0.762	Methylation_HM450K	7647	Mesothelioma
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii		miRNA	203	
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii		RPPA	41	
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii		RNAseq_HiSeq	5852	
			stageiv				
		stageii	stageiii				
			stageiv				
TCGA-SKCM	80	stagei	stageiii	0.350	Methylation_HM450K	7534	Skin Cutaneous
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii		miRNA	218	
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii		RPPA	47	
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii		RNAseq_HiSeq	5691	
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii		Methylation_HM27	4796	
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii				
			stageiv				
			stageiii				

Table 2: Overview of real datasets to benchmark *(continued)*

Dataset	N	Y=0	Y=1	Prop(Y = 1)	Omic	P	Disease
TCGA-STES	25	stageii	stageiv	0.600	miRNA_GA_miR	175	Stomach and Es
		stagei	stageiii				
			stageiv				
		stageii	stageiii		RPPA	42	
			stageiv				
		stagei	stageiii		RNAseq_HiSeq	5590	
			stageiv				
		stageii	stageiii				
			stageiv				
TCGA-THCA	217	stagei	stageiii	0.318	Methylation_HM450K	6367	Thyroid Carcino
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii		miRNA	231	
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii		RPPA	25	
			stageiv				
		stageii	stageiii				
			stageiv				
		stagei	stageiii		RNAseq_HiSeq	5537	
			stageiv				
		stageii	stageiii				
			stageiv				

* Features represents number of predictors in invididual omics

Simulation studies

We examined the methods on controlled setting of simulation studies. Methods were tested against multiomics data with varying signal, correlation, number of samples, and number of predictors on Fig **??**.

Computational time

We observed notable variations in the computation time of most methods with changing dataset sizes (categorized as small if the size of dataset is less than median of all), while DIABLO exhibited consistent performance regardless of dataset size. Conversely, Cooperative Learning displayed a comparatively longer median time than other methods. (Fig 1A).

Real data classification performance

Regarding classification performance, we employed the area under the curve (AUC) score for method comparison across datasets (Fig 1B). Mogonet emerged as the weakest performer, with MOFA + glmnet showing only marginal improvement. In contrast, Cooperative Learning demonstrated the highest performance, followed by DIABLO and RGCCA. Interestingly, DIABLO with a null design outperformed than its full design choice. Nevertheless, all methods performed poorly on the tcga-thca dataset.

Feature performance

We further investigated each method's ability to identify significant biomarkers and examined the overlap between them, calculating the similarity in their feature sets (Fig 1C). Most methods identified similar biomarkers in TCGA datasets but exhibited minimal overlap in the rosmap and gse71669 datasets. Notably, DIABLO and RGCCA tended to select similar biomarkers due to their shared backend framework of canonical correlation analysis.

Discussion

- Explain results, and relevant comments to these

Conclusion

Overall, Cooperative Learning emerged as the top performer in terms of classification performance, disregarding its computational time overhead. Conversely, MOGONET, leveraging deep neural networks, displayed the weakest performance. While acknowledging the potential for further enhancements through exploration of additional datasets and methodologies, we encountered a significant challenge of no publicly available repository/data portal with curated multiomics data. To address this gap, we have begun to curate multiomics studies for various omics data (e.g., transcriptomics, methylation, proteomics) and associated clinical metadata across a range of cancer and non-cancer diseases from various public sources to create the first data portal for multiomics data (~ 100 studies) in both standard formats mentioned such that they can easily be used by other researchers. We anticipate this exercise to be highly relevant to the community utilizing multiomics data for their research, such will help inform the development of a new integrative method that is applicable to different types (bulk, single-cell, or spatial) of multiomics data with the extensibility of our proposed MESSI pipeline.

Methods

Metric

First, defined the metrics used in benchmark analysis ...

- TP: true positive
- TN: true negative
- FP: false positive
- FN: false negative

wheras, it could be constructed to the following:

$$\text{accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Method Algorithms

We present here the mathematical details of the integration methods benchmarked ...

DIABLO

DIABLO is ...

Multiview

Multiview is ...

MOFA

MOFA is ...

MOGONET

MOGONET is ...

RGCCA

RGCCA is ...

TODO

- Explain what is the data input in mathematical form
- Then introduce the integration methods used here
- Each should have their unique characterisitic, and what problem it solves in math, with proper citation
- Then on how data was simulated, hopefully add math details
- Then explain what metrics to be used in both simulated data and real datasets
  - Define what is TP, FP, FN, TN
  - and the the actual formulas of the metrics

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More information

This will be Appendix A.

One more thing

This will be Appendix B.